Measuring the Cognitive Load of Supervising Parallel AI Coding Agents: A Local, Research-Grounded, Individualized Index

Marinos Prevenios
Independent Research github.com/sagium/ai-code-cognitive-stress

Abstract

Agentic coding tools have placed software developers in a role that was previously confined to specialist operators: one human concurrently supervising several semi-autonomous agents at machine pace, interleaving prompts, judging outputs, and recovering interrupted trains of thought all day. This is a distinct cognitive regime, it accumulates, and existing developer-productivity dashboards do not measure it—they count output, not cognitive cost. We present a method and an open, *local-only* tool that turns the session logs these tools already generate into a daily, individualized load index. It is meant as a self-assessment aid for an individual engineer—a good-enough proxy for the multitasking overload that precedes burnout—and it deliberately scopes to interaction with coding agents, not the entire workday. The index decomposes into three behaviourally grounded axes—Concurrent Operational Demand Load (CODL), an Interruption Index, and a Closure Deficit—each anchored to an established literature (working-memory capacity, supervisory-control fan-out, the interruption and attention-residue literature, and the Job Demands–Resources model). A central refinement weights concurrency by *engagement*: a session counts at full load only while actively driven and at a discounted weight while "cooking" in the background, a distinction we ground in prospective-memory monitoring costs and the cognition of unfulfilled goals. We deliberately frame the work as a *taskload* measurement (objective demand from events) rather than *workload* (felt experience), state every assumption that is currently unvalidated, and devote a substantial part of the paper to challenging our own construct. We close with falsifiable predictions and a concrete validation roadmap intended to let the community confirm, calibrate, or refute the index.

1 Introduction

Large-language-model coding assistants—Claude Code, OpenAI Codex CLI, Aider, and others—are routinely run several at a time, or in many concurrent sessions of one. The human operator no longer types most of the code; they *supervise*: dispatching tasks, context-switching between agents, judging partial results, and re-engaging stalled threads. The closest well-studied analogue is supervisory control of multiple semi-autonomous vehicles, where a single operator's performance is known to collapse non-linearly past a personal "fan-out" limit, often *before* the operator consciously registers overload [4,6,25].

Three properties make this load hard to see from the inside. First, it is *concurrent*: holding multiple open agent conversations taxes working memory, whose capacity for independent items is famously small (about four) [3]. Second, it is *interruption-dense*: switching between agents and reacting to failures fragments attention and leaves residue that slows the resumed task [13–15,32]. Third, it is *closure-poor*: agentic work spawns many loops that stay open, and demands that exceed recoverable resources are the mechanism of burnout in the Job Demands–Resources model [7]. Chronic load without recovery—not isolated peaks—is what damages people over time [18,27].

This is not a speculative concern. For AI-assisted coding specifically, a growing body of peer-reviewed work finds that effort does not disappear under assistance but *shifts* from writing to verification and oversight: programmers spend roughly half their session time in assistant-specific states—verifying suggestions, deferring thought, prompt-crafting, and editing [19]; developers given an AI assistant write less secure code while being *more* confident it is secure [21]; and professional engineers show automation complacency and rising reliance over the course of a single task [23]. A mixed-methods survey of 442 developers links GenAI adoption to burnout through intensified job demands [10]. Crucially, this load is hard to read from the inside: in a randomized trial, experienced developers were slowed by $\approx 19\%$ when allowed AI tooling yet believed it had sped them up [2]—a perception–reality gap echoed in longitudinal telemetry [24] and large-scale developer surveys [29]. A self-perception this miscalibrated is precisely what an honest, behavioural, after-the-fact picture is for.

Our goal is modest and concrete: to give an individual developer an honest, research-grounded, *private* picture of this load over time, scored against *their own* historical baseline rather than a population norm. We do not claim to measure stress, to diagnose burnout, or to replace

validated subjective instruments. We claim to make a previously invisible *behavioural* quantity visible, on the user's own machine, with every threshold traceable to the literature.

The intended use is *individual self-assessment*: a working engineer gauging their own multitasking and exhaustion over time, with the index as a good-enough behavioural proxy for the overload patterns that precede burnout—an early warning to act on, not a clinical diagnosis (§6). A scope condition follows from how the signal is built: it sees only the developer's *interaction with coding agents*, not the whole working day. That partial coverage is itself informative. A day whose agent-interaction load is extreme implies one of two things about the rest of that day—either little cognitive capacity remains for other meaningful work, or that work, including the AI-assisted development itself, is degraded by the attention deficit the index is registering. The index is thus better read as *how much of a finite attention budget agent-supervision consumed* than as a tally of activity.

The contributions are:

1. A decomposition of agent-supervision taskload into three axes with explicit, citation-anchored thresholds (§3).
2. An *engagement-weighted* concurrency measure that distinguishes actively-supervised sessions from background "cooking" ones, grounded in prospective-memory and unfulfilled-goal research (§3.3).
3. An open, dependency-free, local-only reference implementation that serves as a working proof of concept [22] (§4).
4. A deliberately adversarial treatment of our own construct (§6) and a validation roadmap with falsifiable predictions (§7).

2 Background and Related Work

Concurrency and working memory. Cowan's reconsideration places the capacity of focal attention for independent chunks at roughly four [3]; we use this as the reference point past which tracking parallel agents should degrade.

Supervisory control and fan-out. The human-supervisory-control tradition models an operator dividing attention across automated processes [25]. "Fan-out" is the number of robots an operator can effectively manage; it is bounded by *neglect time* (how long a process runs unattended) and *interaction time*, and an agent's *attention demand* is the fraction of operator time it consumes [4, 6]. This fraction framing directly motivates our engagement weighting (§3.3).

Interruption and attention residue. Interrupted work is completed faster but at the cost of higher stress and effort [15]; fragmented work and frequent external switches are costly [14], with residue from an unfinished task impairing the next [13], and cross-modality or cross-tool switches costlier than within-tool ones [32].

Demands, recovery, and the optimum. The Job Demands–Resources model frames burnout as demands outstripping recoverable resources [7]; allostatic-load theory stresses that *sustained* activation, not peaks, causes harm [18]; and psychological detachment in off-hours predicts next-day well-being [27, 28]. Performance follows an inverted-U in arousal/load [33], the "flow channel" between boredom and anxiety [5], motivating a *personal optimum* rather than a fixed ceiling.

Open loops and monitoring costs. Two literatures ground the claim that a backgrounded task is neither free nor full cost. Unfulfilled goals remain active in memory and intrude on unrelated tasks until the goal is closed or a concrete plan is made [16]; and maintaining a pending intention while doing other work measurably slows that ongoing task—the prospective-memory "monitoring cost," on the order of 15–20% of baseline [26].

AI-assisted coding and developer cognition. A young but fast-growing empirical literature studies these tools directly, and it is what makes the present concern concrete rather than analogical. AI assistance shifts effort toward verification: programmers spend ≈51.5% of session time in suggestion-specific states [19], and merely *informing* developers that code is LLM-generated raises their measured NASA-TLX workload [30]. Over-reliance is demonstrated rather than hypothesised—a "false sense of security" among assistant users who write less secure but more confidently-trusted code [21], and automation complacency among professional engineers [23]—and the interaction is bimodal, alternating fast *acceleration* with slow, validation-heavy *exploration* [1]. Burnout now has its first peer-reviewed anchor in a Job Demands–Resources analysis of GenAI adoption [10]. What this literature still largely lacks is direct, longitudinal, AI-vs-no-AI measurement of cognitive load in everyday professional work [24]; our index approaches that gap from the opposite direction—an objective behavioural signal computed continuously from real sessions on the developer's own machine, rather than a one-off laboratory measurement.

What we are *not*. The validated instrument for subjective workload is NASA-TLX [12]; for diagnosed burnout it is the Maslach Burnout Inventory [17]. Our index is neither; it is an objective behavioural proxy, and §6 treats the gap as the central threat.

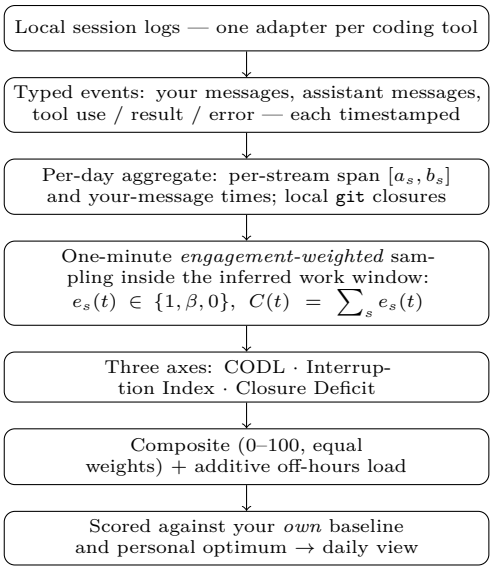

Figure 1: From logs to a daily score, entirely on the user's machine. Overlapping sessions are swept at one-minute resolution and weighted by engagement (foreground 1, background β), so raw concurrency becomes the engagement-weighted load $C(t)$ (§3.3); the three axes and composite are evaluated per day and read only against the individual's own history.

3 Method

The pipeline from raw logs to a daily score is summarised in Figure 1; the remainder of this section defines each stage.

3.1 Data model

Input is the set of local session transcripts a coding tool already writes, plus local `git` history. A pluggable adapter maps each tool's log into a common vocabulary of typed events (user message, assistant message, tool use, tool result/error) with UTC timestamps. Events are reduced to per-day *stream* records: one stream is one session on one day, summarised by its first/last event timestamps $[a_s, b_s]$, per-type counts, and the multiset of the user's own message timestamps U_s. All processing is local; nothing is transmitted (§4).

3.2 Work window

Axes are computed inside a daily work window $W = [w_0, w_1]$ *inferred per operator*: $w_0 = \lfloor p_{10} \rfloor$ and $w_1 = \lceil p_{90} \rceil$ are the p_{10} and p_{90} of the local-time hours at which the operator sends messages, floored and ceiled *outward* to whole hours so the band is never narrower than the percentile span, pooled over the baseline window and applied as one stable band to *every* calendar day (a degenerate or empty band falls back to the cold-start default below). The window is therefore individual to the operator rather than a property of the experiment. We make no assumption about which days are working days—there is no privileged weekend; a person can work any day. "Work" is whatever falls inside the operator's own inferred hours, on any date; activity outside those hours is the off-hours *recovery* signal, again on any date. A manual band may be pinned in configuration to override inference; before enough data accrues (< 5 distinct dates of activity) a conventional 09:00–19:00 local band serves as a cold-start default. We sample the window at one-minute resolution; let T be the set of sample instants.

Off-hours is thus defined relative to the operator's own agent-interaction hours, not the calendar: the index measures the *additional* strain of agentic coding, not total workload or work–life balance. It follows that the index has no notion of a *rest day*—a seven-day-a-week operator with consistent hours registers no off-hours signal—which is a scope boundary rather than a defect.

3.3 Engagement-weighted CODL

The naive concurrency count treats a stream as one continuously-active locus across its whole $[a_s, b_s]$ span. This over-counts: a session left running while the operator is away reads identically to one being actively juggled. We instead weight each stream by *engagement* at each instant. Let g be a foreground grace window (default 5 minutes) and $\beta \in [0, 1]$ a background weight. The instantaneous weight of stream s at time t is

$$e_s(t) = \begin{cases} 1 & \text{if } \exists\, u \in U_s :\ 0 \le t - u \le g \\ \beta & \text{else if } a_s \le t \le b_s \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

That is, a session counts fully only within g of one of the operator's own messages (*foreground*, actively driven); otherwise, while alive, it is "cooking" in the background at weight β. Weighted concurrency and the two CODL summaries are

$$C(t) = \sum_s e_s(t), \quad \overline{\text{CODL}} = \frac{1}{|T|} \sum_{t \in T} C(t), \quad \text{CODL}^\star = \max_t C(t). \tag{2}$$

We retain a raw headcount $H(t) = \sum_s \mathbf{1}[a_s \le t \le b_s]$ and its peak $\max_t H(t)$ as a descriptive "sessions open at once," but the *scored* quantities use the weighted form.

Choosing β. A background locus is neither free nor full cost. A session left "cooking" while the operator attends elsewhere is precisely the *out-of-the-loop* regime of human supervisory control: the operator periodically lets a semi-autonomous process run unattended and must re-engage to judge or correct it. That literature is consistent on three points that bear directly on β. Monitoring a process you have let run is not idle time but *effortful*, and

sustained monitoring is itself a source of stress [31]; remaining out of the loop degrades situation awareness and imposes a cost when control is resumed [8, 9]; and under concurrent load operators systematically under-monitor automation (complacency), so the divided-attention cost is real rather than hypothetical [20]. Together these ground a positive residual cost—$\beta > 0$—in a regime far closer to ours than the laboratory. For the *magnitude* we still borrow a number: the prospective-memory cost of holding a pending intention, ≈15–20% of ongoing-task capacity [26], with unfulfilled goals consuming working memory until closed [16], and the supervisory-control "attention demand" fraction on the operator side [4]. We set $\beta = 0.25$, the conservative top of that band, erring toward a stronger anti-fire-and-forget signal. β and g are configuration parameters, not hard-coded constants, so the assumption is auditable and tunable (§7).

3.4 Interruption Index

Tool calls *within* an active session are not counted: while an agent works, the operator is in a "waiting" state in which attention can legitimately detach, so counting each call would inflate the rate by orders of magnitude and violate the standard definition of an interruption as an unscheduled event demanding immediate attention. Three events do pull attention: a tool *error* (the operator may need to intervene), a *cross-stream start* (a new session lit up while another was active, forcing a switch), and a *git rework* event—an amend, rebase, reset, or cherry-pick read from the reflog (§3.5), where a history rewrite reopens a loop thought closed and is thus a self-interruption with attention residue [13]. Rework events collapse to one count per logical operation (a rebase's terminal entry, not one per replayed commit). With E^{win} the tool-error count apportioned to the work window, X the in-window cross-stream starts, R the in-window rework count, and h the window length in hours,

$$I = \frac{1.5\,E^{\text{win}} + 3.0\,X + 2.0\,R}{h}. \tag{3}$$

The weights follow the relative cost of external versus self-interruption in the interruption literature [14,15,32]—rework sits between a single failed call and a full cross-stream switch; the normalization ceiling is 10/hour, well above the field baseline of a working-sphere switch roughly every ≈11 minutes (≈5/hour) observed in knowledge work [11].

3.5 Closure Deficit

An open loop—a task dispatched but not yet resolved—keeps consuming the cognitive resource until it is closed or concretely planned [16]; closure removes the residue an unfinished switch leaves on the next task [13] and is itself a recovery resource [28]. We therefore measure the share of the day's git-visible *opened* loops that were never *closed*. A loop is opened when a stream starts inside the work window. It is *closed* by a real closure event—a `git` *push*, *commit*, or *merge* that is the *operator's own*—made in *that session's own repository* and timestamped within the loop's active span plus a short grace tail, $[a_s,\ b_s + g]$; we use $g = 30$ min, since a closure typically lands shortly after the agent falls quiet. The strongest signal is the push: it means the work left the machine, and it is self-scoped by construction (only the operator's own pushes write an `update by push` entry to this clone's remote-tracking reflog, so a server-side merge bot cannot forge one). Commits and merges are scoped to the operator's own author identities—the set of `user.email`/`user.name` values configured in local `git` (global plus each scanned repository, which already captures an operator who commits under several accounts). Without this scope, in a shared monorepo the closure signal collapses into *"was anyone on the team committing while I had a session open"*: in our own data the dominant repository carried 4,477 commits from 48 authors (1,051 from a merge bot) and *none* authored by the operator under the all-author rule, so teammates' and the bot's commits spuriously closed the operator's loops and understated the deficit. Each closure event closes at most one loop, matched greedily to the earliest-ending open loop it covers, so a single commit cannot mask several abandoned loops. The repositories scanned are, by default, auto-discovered from the working directories the sessions themselves recorded: each session's `cwd` is walked up to its nearest `.git` root, so the closure signal is on out of the box without the operator hand-listing repositories. This stays inside the local-only invariant—it reads cwds the sessions already logged and runs read-only local `git`, walking no further than those directories.

Closures are attributed *per repository and per author*: a commit in one project can close only loops opened by sessions that ran in that same repository, and only when the operator authored it, so neither cross-project nor cross-author activity can spuriously close a loop. A loop counts toward the deficit only when `git` can speak to it—when its repository saw at least one of the *operator's own* events that day. Loops in a repository `git` never touched by the operator that day (or whose `cwd` resolves to no tracked repository) are *excluded by design*, so the axis scores closure only over the loops it can actually observe, rather than penalising sessions—investigation, review, planning—that were never meant to produce a commit. History-*rewriting* operations (amend, rebase, reset, cherry-pick)—read from the `git` reflog, the only place they are recorded—make a repository git-visible but are *not* themselves closures: a rewrite reopens or churns a loop thought closed, so a rebased-but-uncommitted loop stays *unclosed*, and the rewrite is routed to the Interruption Index as rework (§3.4) rather than to this axis. With $\mathcal{O} = \{s : a_s \in W\}$ the loops opened in the window, $\mathcal{C} \subseteq \mathcal{O}$ the *correlatable* loops—those whose repository was `git`-active that day—and k the number

of them closed by the greedy repository-and-time match,

$$\text{ClosureDeficit} = \text{clip}\Big(1 - \frac{k}{|\mathcal{C}|},\, 0,\, 1\Big), \quad (|\mathcal{C}| > 0), \quad (4)$$

and *undefined* when no loop is correlatable ($|\mathcal{C}| = 0$). The Closure Deficit has meaning only on git repositories, so a day git cannot speak to—debugging, generic chat, or work in a repository the operator did not touch—is *omitted as data*, not scored. The distinction is load-bearing: 0 means "you opened loops and closed them all" (genuine perfect closure), whereas an omitted day means "there was no git work to assess." Collapsing the second into a 0 would credit a pure-debugging day with perfect closure and dilute its composite, so the two are kept distinct: the omitted day contributes to neither the axis nor any rollup average, and the composite renormalises over the axes that remain (§3.6). A day that opens many loops and closes few is exactly the demands-without-recoverable-resources pattern the JD-R model identifies with burnout [7].

Crucially, this axis is built from per-session *correlation* of opened and closed loops, not from the concurrency time-series $C(t)$, so it is independent of the CODL shape by construction: two days with identical $C(t)$ (hence identical $\overline{\text{CODL}}$) receive different deficits if one committed its loops and the other left them open—information $C(t)$ cannot carry. This removes the collinearity of earlier versions, in which the deficit was the fraction of samples with $C(t) > 1$ and thus a re-expression of concurrency. The remaining proxy is the correlation itself: a closure is matched to a session by repository, author, and time overlap, not by a shared identifier (session transcripts and commits carry none), so an own-authored commit may still be matched to a different *own* same-repo session active in the same window (§6). The earlier all-author version had a far larger confound—a *teammate's* commit closing the operator's loop in a shared repository—which author-scoping removes. With no git repository configured at all the axis has no basis to exist, so it is omitted entirely (an earlier version fell back to a $C(t) > 1$ concurrency-presence proxy here; that proxy was *removed*—it was not a git signal and re-expressed concurrency the axis is meant to be orthogonal to).

3.6 Composite

Each available axis is normalized to $[0, 1]$ and blended with equal weights. When the Closure Deficit is defined for the day, all three contribute:

$$\text{Composite} = \frac{100}{3}\Bigg(\min\Big(1, \tfrac{\overline{\text{CODL}}}{5}\Big) + \min\Big(1, \tfrac{I}{10}\Big) + \text{clip}(\text{ClosureDeficit}, 0, 1)\Bigg). \quad (5)$$

On a day with no git-correlatable activity the Closure Deficit is undefined (§3.5); the blend then *drops* that axis and renormalises over the remaining two—its weight is redistributed to CODL and Interruption, not imputed as a perfect-closure 0. So a heavy debugging or chat day is scored on the load and interruption it actually carried rather than discounted for closing loops it never opened. In general the composite is the weight-normalised mean over whichever axes are defined for the day. The CODL ceiling is set to 5—one above Cowan's ≈ 4 capacity limit—so a day operating *at* the working-memory boundary scores 0.8 (the warning band) rather than saturating the axis at 1.0. Off-hours work then adds an *additive* load to this base. Let o be the day's off-hours *engaged* minutes—one-minute instants outside W during which the operator was actively driving a session, i.e. within the foreground grace g of one of their own messages (§3.3). The reported composite is

$$\text{Composite}^{*} = \min\big(100,\ \text{Composite} + 30 \cdot \min(1, o/90)\big). \quad (6)$$

Two design choices matter here. First, the toll is *additive*, not multiplicative: off-hours work always counts, so a day worked *entirely* outside the window—zero in-window base—still scores a positive composite, which a multiplicative form would wrongly leave at zero. Second, o is anchored to *interaction*, not stream liveness: a background session left running while the operator is away contributes nothing, and `git` commits never enter—only the operator actually being present and driving the agent outside their own hours counts (e.g. waking in the night to check on a run). The inferred band W is the operator's norm; engaged work outside it is treated as a deviation from that norm. Like the background weight β, the magnitude (+30 points, saturating at 90 engaged minutes) is an explicit prior, not a fitted value (§6), grounded in disengagement as a recovery resource [28]. Equal weights are the explicit *null hypothesis* for this version: we have no evidence that one axis dominates felt load, and we prefer to state that than to fit weights to a single user's data. Replacing equal weights with empirically calibrated ones is a primary validation target (§7). The reference implementation colors and ranks each day against the operator's *own* percentile distribution (p50/p75/p90) computed over active days in the baseline window—never a population norm—reinforcing that only relative movement within one person's history is interpreted (see the corresponding caveat in §6).

3.7 Personal optimum and recovery

Performance follows an inverted-U with load [5, 33]. We bucket historical workdays by $\overline{\text{CODL}}$ (width 0.5) and score each bucket by closure achieved against off-hours follow-up; the midpoint of the best bucket is reported as the operator's individual "flow channel" target, a line on the chart rather than a ceiling. Because the Closure Deficit is now independent of concurrency (§3.5), this scoring term is no longer a function of the very $\overline{\text{CODL}}$ used to form the buckets, so the optimum is

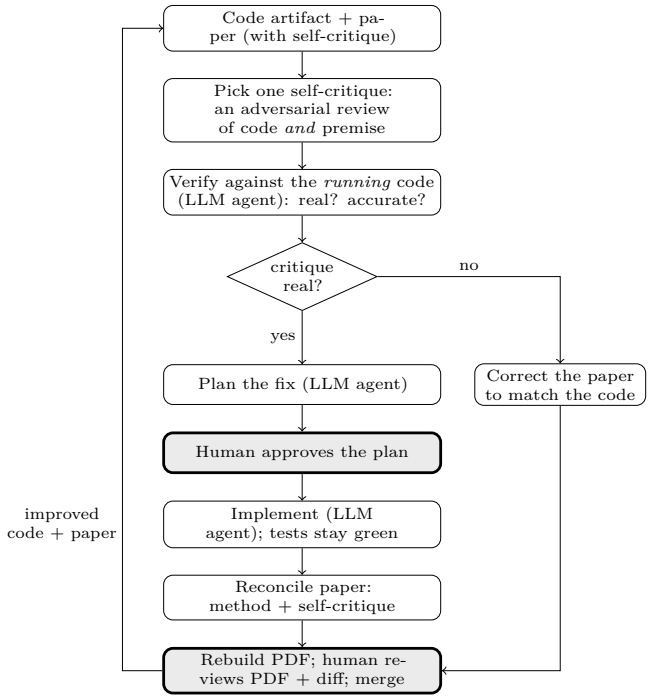

Figure 2: The `reconcile-critique` loop (§3.8): the paper's self-critique drives revision of an *executable* artifact. Two human gates (shaded) bracket every change—plan approval and merge review. Verification and planning use one LLM agent; implementation a separate one. A pass that finds the critique unfounded corrects the paper instead of the code.

no longer circular: it asks which load band *actually* closed the most loops, rather than rediscovering the load band with the least concurrency. It requires at least 14 active days to stabilise, below which the report shows "calibrating." Off-hours activity (on any date) is flagged because sustained load without detachment, not peaks, predicts harm [18, 27, 28].

3.8 Method revision: the paper as adversarial reviewer

The method above is not static; it is the current state of a deliberate revision loop (Figure 2) in which the paper and the implementation co-evolve under human direction. The tool was built first, as the concrete artifact this paper documents (§4). Writing the paper then forced each assumption to be stated precisely enough to attack, and the resulting self-critique (§6) became the *driver* of subsequent change rather than a closing disclaimer: each threat is read as an adversarial review of the code and of the premise behind an axis.

A single pass takes one self-critique and first *verifies it against the running code*—does the weakness actually exist, and is it described accurately?—then either corrects the paper where it mis-states its own artifact, or changes the method and code to remove the weakness and rewrites the affected §3 and §6 text to match, or both. The work is carried out with agentic coding assistants—the same class of tool this paper measures—under a human-in-the-loop gate: an agent investigates and proposes a plan; the human approves it before any code or prose is touched; an agent implements; the test suite must stay green; the paper is rebuilt; and the human reviews the rendered result and the diff before the change is accepted. Several axes reached their current form this way—the per-operator inferred work window, the interaction-anchored off-hours load, the removal of the hard-coded weekend, and the in-domain grounding of the background weight β each began as a §6 critique.

Two properties make this more than ordinary editing. First, the artifact is *executable*, so a critique of the premise is checkable: if the paper claims the code does X, that is either true of the running code or it is a defect in one of them, and the loop must resolve the disagreement rather than soften the wording. Second, the process is *reflexive*—a paper about the cost of supervising agentic coding is itself produced by supervising agentic coding—so the authors are incidentally $n{=}1$ subjects of the regime the index scores (§6). We make the loop explicit because it, more than any single threshold, is the method's route to eventually being right: the index is built to be falsified and revised, and §7 states the external evidence meant to drive later passes.

4 Implementation

The method is realised in an open-source reference implementation [22] that doubles as the proof of concept for this paper: every axis, threshold, and the engagement weighting described here is exactly what the released tool computes.[1] It is pure Python standard library (no third-party runtime dependencies), which keeps the trust surface small and the install trivial. It reads only local files, performs no network I/O at any stage, and writes a self-contained HTML report plus a machine-readable JSON sibling. Per-day aggregates are cached keyed by source-file modification time, so metric or weighting changes recompute from cache without re-parsing logs. Adding a new coding tool is a single adapter implementing the event protocol; the aggregate, metric, optimum, and rendering stages are tool-agnostic. The tool is local-only by design: session transcripts frequently contain proprietary code and secrets, so the generated artefacts are treated as similarly sensitive and never leave the machine.

The implementation also provides two optional, local-only pathways toward the multi-developer calibration of §7. The first is an anonymized, consent-gated *research export*: a per-user file of derived daily metrics (and their

[1] Source and documentation: `https://github.com/sagium/ai-code-cognitive-stress`. The repository is being prepared for public release; the qualitative behaviour described in this section was produced by the released code and is reproducible from a user's own logs.

components) with the timezone dropped and calendar dates randomly shifted, which the operator may choose to upload manually—the tool itself transmits nothing, preserving the local-only invariant. The second is a population *calibration* step that pools such exports and proposes normalization ceilings and composite weights that cover the observed range of work patterns. Both read and write only local files. The calibration is *unsupervised*: it fits the axis *scales* to the population distribution and suggests weights from the redundancy among axes, but it does not fit weights to felt load—that still requires a subjective criterion (§7).

5 Illustrative Example

To make the axes concrete we report one author's own data for a single month (19 active workdays). We stress that this is an *illustration of the instrument*, not evidence of validity; n=1, the subject is an author, and there is no ground-truth load measure here. Qualitatively, the engagement-weighted CODL pulled composite scores down on days dominated by background "cooking" sessions while leaving interruption-dominated days largely unchanged, and the weighted peak CODL* separated a day of genuine four-way *active* supervision from a day with four sessions merely open. This is the behaviour the design intends; whether it tracks *felt* load is exactly the open question of §7.

6 Threats to Validity and Self-Critique

We consider the following the most serious challenges to the method, roughly in order of severity. We see no benefit in understating them.

Taskload is not workload. Every axis is computed from objective event streams; none measures felt experience. Correlation with felt overload is only moderate, and objective and subjective workload are known to dissociate. A calm composite does not prove the operator felt calm, and vice versa. The validated subjective instrument is NASA-TLX [12]; until we correlate against it (or experience sampling), the index is an unvalidated proxy for the construct it names.

The supervisory-control analogy is borrowed, not established. Fan-out, neglect time, and attention demand were developed for UAV/robot operators [4, 6]. Their transfer to LLM oversight is plausible—both involve one human time-sharing across semi-autonomous processes—but unproven. LLM agents differ in failure modes, feedback latency, and the verbal/code modality of interaction. Recent AI-coding studies give the transfer partial, in-domain support: automation complacency and rising reliance have now been observed directly in professional developers using LLM assistants [23], and informing developers that code is LLM-generated measurably raises their NASA-TLX workload [30]. These vary trust and provenance rather than the supervisory fan-out our axes model, so they corroborate the cognitive cost without yet validating the specific fan-out/neglect-time constructs we borrow.

The background weight β is a prior, calibrated only in magnitude. That a background locus carries a positive, effortful, and stress-inducing monitoring cost is supported *in domain* by the supervisory-control and automation literature on out-of-the-loop operation, vigilance, and complacency (§3.3) [8, 9, 20, 31]—a regime far closer to agent supervision than the laboratory. What remains imported is the *magnitude*: the specific 15–20% figure behind $\beta = 0.25$ comes from prospective-memory and goal-priming paradigms [16,26], not from software supervision, so the value is a defensible prior rather than a measurement. The true β may differ by person, task, and tool, and may not be constant within a day; recovering it from data is a validation target (§7).

Equal composite weights are a placeholder. We assert no axis dominates; this is almost certainly false in detail. Without calibration against a criterion, the composite's absolute scale is arbitrary and only its *relative* movement within one person's history is meaningful. The weights are now a configuration parameter rather than a hard-coded constant, and the implementation includes a population step that suggests data-fitted weights from the redundancy among axes across pooled exports (§4). That step is *unsupervised*, however: absent a felt-load criterion it cannot show the fitted weights track subjective load, so equal weights remain the default and the placeholder stands until the supervised calibration of §7.

Closure Deficit correlation is heuristic, and scoped to the operator's own git-visible loops. The axis correlates the operator's own `git` pushes/commits/merges to the loops opened each work window (§3.5), so it is no longer a re-expression of concurrency and the name is honest about what it computes. A closure event closes only a loop in its *own* repository whose active span it falls within (plus a 30 min grace), and closes at most one loop—a commit in repository A can no longer close a loop opened in repository B, which the earlier global count permitted. Two scoping choices matter. First, *author*: commits/merges count only when the operator authored them (matched against the `user.email`/`user.name` identities local `git` records, which covers multiple accounts), and pushes are self-scoped by the reflog that records them. An earlier all-author version had a severe confound—in a shared monorepo a *teammate's* or merge-bot's commit closed the operator's loops, so the axis silently measured team

activity rather than the operator's own closure; on our data it understated the deficit (a pooled 0.25 against a correctly-scoped ≈ 0.54). The honest residual is the inverse: author identity is itself a heuristic, so a commit authored under an un-recorded local identity is missed, and matching on author *name* (not only email) could in principle admit a distinctly-named collaborator who happens to share the operator's configured name. Second, the correlation is still a *time-overlap* heuristic: a closure is matched to a session by repository, author, and time, not by a shared identifier, because session transcripts and commits carry none—so an own-authored commit may still be attributed to a *different own* same-repository session active in the same window, and squashed or batched commits under- or over-count real closures (the squash itself, like other history rewrites, is separately counted as rework on the Interruption Index, §3.4); the grace tail is a fixed constant, not calibrated. By *design*, a loop is scored only when the operator's own `git` activity is visible in its repository that day: loops in repositories the operator did not touch are excluded rather than counted as unclosed, so the axis measures closure over observable loops and does not penalise investigation- or review-only sessions that were never meant to commit. Such a day is *omitted as data*—the axis returns no value, not a 0—so it enters neither the rollup averages nor the personal optimum, and the composite renormalises over the axes that remain rather than crediting the day with a perfect-closure 0 (§3.6); an earlier version that scored these days as 0 silently flattered closure and diluted their stress score. This is deliberately low-power: on our data $\approx 82\%$ of opened loops are dropped this way and the axis is silent on most days, the cost of refusing to speak without evidence. A day of genuine but *uncommitted, unpushed* coding—work opened and abandoned with no `git` trace—therefore reads as no-signal rather than high deficit, and the local push signal is itself lossy because a merged feature branch pruned from the clone takes its remote-tracking reflog with it. A shared session↔commit identifier would replace the time-overlap heuristic, and platform PR/MR events would sharpen the signal further, but the latter requires a network call and is out of scope for a local-only tool.

Engagement is interaction-anchored by design. Foreground is proxied by proximity to the operator's own messages within a grace window g. Silent engagement (reading, thinking, reviewing diffs without typing) leaves no event and is therefore not counted—but this is a scope boundary by construction, not a measurement gap (§3): the index measures the load *driven by agent interaction*, not total cognitive work. The consequence is conservative in the direction that matters. Anchoring to logged interaction can only *understate* the operator's true cognitive load, so the measured agent-supervision load is a lower bound on total load: a high score is a sufficient indicator of overload regardless of whatever silent work goes unrecorded, because if agent supervision alone saturates capacity there is no headroom left for the rest. The one bias in the other direction is that any operator message—including a perfunctory acknowledgement—grants full foreground weight for the whole window, so a trivial message can be over-weighted by up to g. The 5-minute grace is a default, not fitted; like β it is a configurable, auditable parameter rather than a hidden constant, and both the value of engagement weighting and the sensitivity to g are validation targets (§7).

Cross-tool comparability is unestablished. Different tools log at different granularities (e.g. per-keystroke vs. per-turn), so event counts and stream boundaries are not commensurable across adapters. Aggregating or comparing across tools without normalization risks systematic bias.

Single subject and borrowed thresholds. The capacity threshold (≈ 4 [3]) and interruption ceiling (10/hour) are literature values, not fitted to the population of agent-coding developers, who may differ. All reported behaviour is n=1 and from an author. The reference implementation now provides the machinery to move beyond this—an anonymized per-user export and a population-calibration step that refits the ceilings to the pooled distribution (§4)—so what remains is collecting the multi-developer sample, each participant running the local-only tool on their own machine and only derived metrics pooled. Until that sample exists the thresholds stay borrowed and the data stays n=1 (§7).

Modelling and boundary choices. One-minute sampling, the over-approximation of a stream as alive across $[a_s, b_s]$, and timezone handling are all defaults that could materially move scores for shift workers or bursty users. Timezone is always the operator's own local zone and the tool is single-subject by design, so cross-timezone teams are out of scope rather than a threat to this index. The work window is now inferred per operator from the p_{10}–p_{90} of message hours rather than fixed, which removes the hardcoded-band assumption but introduces its own choices—the percentile edges, the hour rounding, and the five-date minimum before inference engages—and assumes that message timing tracks actual working hours; an operator who works silently or messages at odd hours will get a skewed band, and inference falls back to the fixed default during cold start. The additive off-hours load that lets work outside the band raise the composite is likewise an explicit prior (+30 points, saturating at 90 engaged minutes), not a measured penalty; its magnitude could over- or under-state the true cost of off-hours work and is a target for calibration. That load is anchored to *interaction*—engaged minutes within the grace of the operator's own messages—so it ignores background sessions running while the operator is away

and counts only active supervision of the agent. Silent reading or review without typing, and the lack of any "rest day" notion, are therefore out of scope by construction rather than measurement gaps: the index targets the additional strain of *agentic coding* surfaced through interaction, not total work or work–life balance (§3). Additionally, tool errors are apportioned to the work window by assuming they are uniformly distributed across each stream's $[a_s, b_s]$ lifetime—a defensible default since the per-day aggregate carries no per-error timestamps, but one that could misattribute errors from bursty or session-boundary activity.

Construct and causal claims. "Stress" in the project name is aspirational; the composite is a load index. No causal link to burnout outcomes (e.g. MBI [17]) is claimed or shown; the index is offered as a self-assessment proxy for the overload patterns *associated* with burnout risk—an early-warning aid—not a validated burnout measure. Reactivity is also possible: surfacing the score may itself change behaviour. And because the index covers only coding-agent interaction (§3), it is silent on total workload: a low score does not certify a sustainable day, only a light *agent-supervision* load.

7 Falsifiable Predictions and Validation Roadmap

The above critiques are not fatal; they are a research agenda. We state predictions that would, if tested, confirm or refute the design.

1. **Concurrent validity.** Across a sample of agent-coding developers, daily composite should correlate positively with same-day NASA-TLX [12] or ecological momentary assessments of effort. A null or negative correlation falsifies the composite as a load proxy.

2. **Engagement-weighting ablation.** Engagement-weighted CODL should correlate with felt load *better* than the flat headcount. If weighting does not improve the correlation, β and g add complexity without value.

3. **Weight calibration.** Regressing subjective load on the three normalized axes should yield weights that beat the equal-weight null out-of-sample; the fitted weights become the next version's defaults. The implementation already derives *unsupervised*, redundancy-based weight suggestions from pooled exports (§4); this prediction concerns the *supervised* step the tool cannot perform—fitting and validating weights against a felt-load criterion.

4. **Predictive validity for recovery.** Days high in off-hours activity and composite should predict lower next-day vigor/detachment scores [27]; absence of this relationship weakens the recovery framing.

5. β **recovery from data.** With per-event timestamps and a secondary signal (e.g. self-reported "which sessions were you actually watching"), β can be estimated rather than assumed; the estimate's distance from 0.25 quantifies our prior's error.

6. **Sensitivity analysis.** Composite rankings of days should be stable under reasonable perturbations of g, the sampling interval, and the thresholds; instability would indicate over-tuned, non-robust scoring.

7. **Inferred window vs. fixed band.** The per-operator inferred window should predict felt load (or in-flow time) better than the fixed 09:00–19:00 band out-of-sample. If inference does not beat the fixed band, the percentile machinery adds complexity without value and the fixed default should return.

8. **Off-hours load validity.** Days carrying a high off-hours engaged-minutes load should predict lower next-day vigor/detachment [27]; if the load does not track reduced recovery, its magnitude (+30 points, 90-minute saturation) is mis-specified and should be refit or dropped.

9. **Population threshold check.** With multi-developer data, the ≈ 4 capacity ceiling [3] and 10/hour interruption ceiling can be checked and, if warranted, refit to the agent-coding population; large divergence would mean the borrowed values mis-set the axis scales. The calibration step computes exactly this from pooled exports (§4); what remains is collecting the population.

A modest multi-developer study with experience sampling—each participant running the local-only tool on their own machine, with only the derived per-day metrics and subjective ratings pooled for analysis, so no off-machine data path is added to the tool—is now supported end-to-end by the released tooling (the anonymized export and the pooled-calibration step, §4) and is the primary outstanding work; running it would address the majority of the threats in §6, in particular the single-subject and borrowed-threshold limitations, converting this from a principled instrument into a validated one. Folding real closure events (pushes/commits/merges) into the Closure Deficit, previously listed here as future work, is now done and active by default via repositories auto-discovered from session working directories, with closures scoped to the operator's own author identities and pushes (self-scoped by the local reflog), correlated to sessions by repository, author, and time overlap, and history-rewrite operations routed to the Interruption Index as rework (§3.5); the remaining closure-side work is a reliable session↔commit identifier to replace the time-overlap heuristic, and recovering closure on genuinely

uncommitted loops that the git-visibility scope currently excludes.

8 Ethics and Privacy

The tool reads proprietary source and potentially secrets; we therefore treat local-only processing as a hard invariant, not a feature, and any future data-sharing path would require separate, explicit, opt-in consent. The intended use is individual self-reflection. We explicitly warn against managerial or comparative use across people: the index is calibrated to an individual's own history, the construct is unvalidated, and using an unvalidated load proxy for evaluation would be both statistically and ethically unsound.

9 Conclusion

Supervising parallel AI coding agents is a new, accumulating, and largely unmeasured cognitive regime. We have presented a local, privacy-preserving, literature-anchored method to make its *behavioural* footprint visible to the individual who generates it, including an engagement-weighted concurrency measure that distinguishes active supervision from background cooking. We have been deliberate about what the method does not yet establish, and have laid out falsifiable predictions and a validation path. We hope the explicit assumptions and the open implementation make the construct easy to challenge—which is the fastest way to make it, and any tool built on it, more trustworthy.

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